

OpenStax Calculus Volume 2

Section 7.2 Exercises: Calculus of Parametric Curves

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For the following exercises, each set of parametric equations represents a line. Without eliminating the parameter, find the slope of each line.

1. $x = 3 + t$, $y = 1 - t$
2. $x = 8 + 2t$, $y = 1$
3. $x = 4 - 3t$, $y = -2 + 6t$
4. $x = -5t + 7$, $y = 3t - 1$

For the following exercises, determine the slope of the tangent line, then find an equation of the tangent line at the given value of the parameter.

5. $x = 3 \sin t$, $y = 3 \cos t$, $t = \frac{\pi}{4}$
6. $x = \cos t$, $y = 8 \sin t$, $t = \frac{\pi}{2}$
7. $x = 2t$, $y = t^3$, $t = -1$
8. $x = t + \frac{1}{t}$, $y = t - \frac{1}{t}$, $t = 1$
9. $x = \sqrt{t}$, $y = 2t$, $t = 4$

For the following exercises, find all points on the curve that have the given slope.

10. $x = 4 \cos t$, $y = 4 \sin t$, slope = 0.5
11. $x = 2 \cos t$, $y = 8 \sin t$, slope = -1
12. $x = t + \frac{1}{t}$, $y = t - \frac{1}{t}$, slope = 1
13. $x = 2 + \sqrt{t}$, $y = 2 - 4t$, slope = 0

For the following exercises, write the equation of the tangent line in Cartesian coordinates for the given parameter t .

14. $x = e^{\sqrt{t}}$, $y = 1 - \ln(t^2)$, $t = 1$
15. $x = t \ln t$, $y = \sin^2 t$, $t = \frac{\pi}{4}$

16. $x = e^t$, $y = (t - 1)^2$, at $(1, 1)$
17. For $x = \sin(2t)$ and $y = 2 \sin t$, where $0 \leq t < 2\pi$, find all values of t at which a horizontal tangent line exists.
18. For $x = \sin(2t)$ and $y = 2 \sin t$, where $0 \leq t < 2\pi$, find all values of t at which a vertical tangent line exists.
19. Find all points on the curve $x = 4 \sin t$, $y = 4 \cos t$ that have slope 0.5.
20. Find $\frac{dy}{dx}$ for $x = \sin(t)$, $y = \cos(t)$.
21. Find an equation of the tangent line to $x = \sin(t)$, $y = \cos(t)$ at $t = \frac{\pi}{4}$.
22. For the curve $x = 4t$, $y = 3t - 2$, find the slope and concavity of the curve at $t = 3$.
23. For the parametric curve whose equation is $x = 4 \cos \theta$, $y = 4 \sin \theta$, find the slope and concavity of the curve at $\theta = \frac{\pi}{4}$.
24. Find the slope and concavity for the curve whose equation is $x = 2 + \sec \theta$, $y = 1 + 2 \tan \theta$ at $\theta = \frac{\pi}{6}$.
25. Find all points on the curve $x = t + 4$, $y = t^3 - 3t$ at which there are vertical and horizontal tangents.
26. Find all points on the curve $x = \sec \theta$, $y = \tan \theta$ at which horizontal and vertical tangents exist.

For the following exercises, find $\frac{d^2y}{dx^2}$.

27. $x = t^4 - 1$, $y = t - t^2$
28. $x = \sin(\pi t)$, $y = \cos(\pi t)$
29. $x = e^{-t}$, $y = te^{2t}$

For the following exercises, find points on the curve at which the tangent line is horizontal or vertical.

30. $x = t(t^2 - 3)$, $y = 3(t^2 - 3)$
31. $x = \frac{3t}{1+t^3}$, $y = \frac{3t^2}{1+t^3}$

For the following exercises, find $\frac{dy}{dx}$ at the value of the parameter.

32. $x = \cos t$, $y = \sin t$, $t = \frac{3\pi}{4}$
33. $x = \sqrt{t}$, $y = 2t + 4$, $t = 9$
34. $x = 4 \cos(2\pi s)$, $y = 3 \sin(2\pi s)$, $s = -\frac{1}{4}$

For the following exercises, find $\frac{d^2y}{dx^2}$ at the given point without eliminating the parameter.

35. $x = \frac{1}{2}t^2$, $y = \frac{1}{3}t^3$, $t = 2$
36. $x = \sqrt{t}$, $y = 2t + 4$, $t = 1$
37. Find t intervals on which the curve $x = 3t^2$, $y = t^3 - t$ is concave up as well as concave down.

38. Determine the concavity of the curve $x = 2t + \ln t$, $y = 2t - \ln t$.
39. Sketch and find the area under one arch of the cycloid $x = r(\theta - \sin \theta)$, $y = r(1 - \cos \theta)$.
40. Find the area bounded by the curve $x = \cos t$, $y = e^t$, $0 \leq t \leq \frac{\pi}{2}$ and the lines $y = 1$ and $x = 0$.
41. Find the area enclosed by the ellipse $x = a \cos \theta$, $y = b \sin \theta$, $0 \leq \theta < 2\pi$.
42. Find the area of the region bounded by $x = 2 \sin^2 \theta$, $y = 2 \sin^2 \theta \tan \theta$, for $0 \leq \theta \leq \frac{\pi}{2}$.

For the following exercises, find the area of the regions bounded by the parametric curves and the indicated values of the parameter.

43. $x = 2 \cot \theta$, $y = 2 \sin^2 \theta$, $0 \leq \theta \leq \pi$
44. [T] $x = 2a \cos t - a \cos(2t)$, $y = 2a \sin t - a \sin(2t)$, $0 \leq t < 2\pi$
45. [T] $x = a \sin(2t)$, $y = b \sin t$, $0 \leq t < 2\pi$ (the “hourglass”)
46. [T] $x = 2a \cos t - a \sin(2t)$, $y = b \sin t$, $0 \leq t < 2\pi$ (the “teardrop”)

For the following exercises, find the arc length of the curve on the indicated interval of the parameter.

47. $x = 4t + 3$, $y = 3t - 2$, $0 \leq t \leq 2$
48. $x = \frac{1}{3}t^3$, $y = \frac{1}{2}t^2$, $0 \leq t \leq 1$
49. $x = \cos(2t)$, $y = \sin(2t)$, $0 \leq t \leq \frac{\pi}{2}$
50. $x = 1 + t^2$, $y = (1 + t)^3$, $0 \leq t \leq 1$
51. $x = e^t \cos t$, $y = e^t \sin t$, $0 \leq t \leq \frac{\pi}{2}$ (Use a CAS for this and express the answer as a decimal rounded to three places.)
52. $x = a \cos^3 \theta$, $y = a \sin^3 \theta$ on the interval $[0, 2\pi)$ (the hypocycloid)
53. Find the length of one arch of the cycloid $x = 4(t - \sin t)$, $y = 4(1 - \cos t)$.
54. Find the distance traveled by a particle with position (x, y) as t varies in the given time interval:
 $x = \sin^2 t$, $y = \cos^2 t$, $0 \leq t \leq 3\pi$.
55. Find the length of one arch of the cycloid $x = \theta - \sin \theta$, $y = 1 - \cos \theta$.
56. Show that the total length of the ellipse $x = 4 \sin \theta$, $y = 3 \cos \theta$ is $L = 16 \int_0^{\pi/2} \sqrt{1 - e^2 \sin^2 \theta} d\theta$, where $e = \frac{c}{a}$ and $c = \sqrt{a^2 - b^2}$.
57. Find the length of the curve $x = e^t - t$, $y = 4e^{t/2}$, $-8 \leq t \leq 3$.

For the following exercises, find the area of the surface obtained by rotating the given curve about the x -axis.

58. $x = t^3$, $y = t^2$, $0 \leq t \leq 1$
59. $x = a \cos^3 \theta$, $y = a \sin^3 \theta$, $0 \leq \theta \leq \frac{\pi}{2}$
60. [T] Use a CAS to find the area of the surface generated by rotating $x = t + t^3$, $y = t - \frac{1}{t^2}$, $1 \leq t \leq 2$ about the x -axis. (Answer to three decimal places.)
61. Find the surface area obtained by rotating $x = 3t^2$, $y = 2t^3$, $0 \leq t \leq 5$ about the y -axis.

62. Find the area of the surface generated by revolving $x = t^2$, $y = 2t$, $0 \leq t \leq 4$ about the x -axis.
63. Find the surface area generated by revolving $x = t^2$, $y = 2t^2$, $0 \leq t \leq 1$ about the y -axis.